

Data for the UDef-ARP (or UDef-A within TerrSet) exercises are provided with the TerrSet Tutorial data, which can be downloaded from <https://doi.org/10.5281/zenodo.1420200>.

The download link for the data for just the UDef-ARP <https://zenodo.org/records/20073844/files/TerrSet-liberaGIS-Tutorial-UDefA.zip?download=1> at the time of this writing. However, if that data gets updated, the link may change. In that case, look in the permanent link to the TerrSet Tutorial data mentioned above, and scroll down to the UDef-A section.

Most tutorial datasets are small in size, so that you can test something out without needing to use much disk space. UDef-ARP is designed to only work with datasets that are the size of REDD jurisdiction sized maps, which are the size of some countries or large provinces. This tutorial dataset is 14.5 GB, and you will need more disk space as well for the maps that will be created at the various stages.

This tutorial is using TerrSet / Idrisi RST (.rst) image files. The same steps could be done with TIFF (.tif) files.

UDEF-ARP EXERCISES

INTRODUCTION

In 2023, Verra (www.terra.org) introduced a new methodology (VT0007) entitled *Unplanned Deforestation Allocation Risk Modeling and Mapping Procedure (UDef-ARP)*. The methodology was developed by Clark Labs in partnership with TerraCarbon (www.terracarbon.com) for Verra to allocate (map) the risk of unplanned deforestation in order that REDD (Reducing Emissions from Deforestation and Forest Degradation) projects could establish baselines.

BASELINES

A baseline is the quantity of deforestation expected within a project area that is expected to occur if project interventions were not implemented. Baselines are critical as they set the maximum expected income a project can receive from project investors, if they are successful in preventing unplanned deforestation.

RISK MAPS

The requirement of the risk map developed is very different from the scenario-based predictions normally associated with land change modelling. The issue isn't which exact pixels will deforest, but rather, within any project boundary, what is the expected quantity of deforestation. UDef-ARP does this by assigning each pixel an expected density of deforestation, expressed in hectares per pixel per year. Verra thus determines project baselines by summing the densities within each project's boundary.

The units of these densities at the pixel level are typically quite small. For example, if one is working with 30m imagery, each pixel is 30m x 30m = 0.09 ha. If the assessed probability of deforestation for such a pixel is 10%, then the expected density is 0.009 ha/pixel/yr.

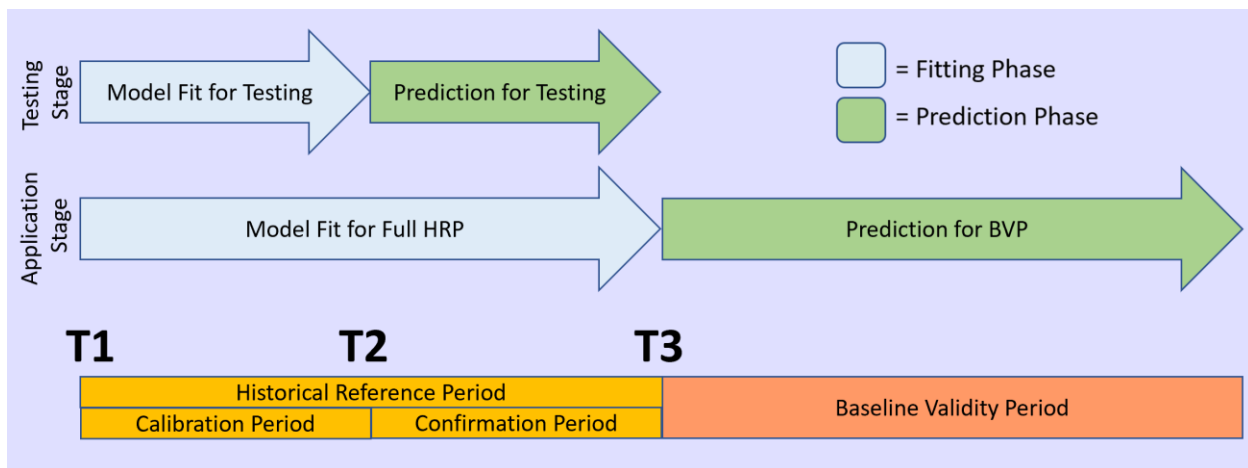
Under Verra's guidelines, all REDD projects within a jurisdiction must use the same risk map. A jurisdiction is typically either a whole country, or some major division of a country, such as a province or state. The intention of mapping at the jurisdiction level is to facilitate the alignment of individual projects with national climate goals.

RISK MAP DEVELOPMENT

Risk maps are empirically developed. That is, they look at deforestation over a historical reference period along with covariates (variables that covary with the deforestation experienced) and develop a model that can both explain (fit) the historical past and predict the future (assuming that the future behaves like the past). Verra currently defines a historical reference period (HRP) as 10 recent years, and it is used in two ways: testing and application.

THE TESTING STAGE

The first way is to develop and test models. For this, the HRP is split into two periods: a calibration period (CAL) and a confirmation period (CNF) as illustrated below: The CAL and CNF are each 5 years in length.



In the CAL, a map of deforestation from time T1 to time T2 is used as the dependent variable. Together with maps of covariates appropriate to T1, models are empirically developed and compared with a goodness of fit test. Then these models are applied to the CNF by updating the covariates for T2 and producing a prediction of deforestation from T2 to T3. This is then compared to the actual deforestation over the CNF to test prediction skill.

Among the models tested, one is a required benchmark model – a simple empirical procedure that sets a minimum standard of quality. Ultimately, the final model chosen must demonstrate a better quality of fit in the CAL and a better prediction skill in the CNF than the benchmark.

THE APPLICATION PHASE

Having chosen a final modelling approach, the application stage is begun, where the same modeling procedure is repeated using the entire HRP (from T1 to T3) to develop a new model. This new model is then applied to updated data from T3 to predict the baseline for a 5-10 year validity period.

EXERCISE 5-1: THE CASE STUDY DATA: ACRE, BRAZIL

To illustrate the use of UDef-ARP, we consider the case of the State of Acre, Brazil. Acre is in the western Amazon and has experienced recent frontier deforestation, primarily for the expansion of cattle ranching.

- A. The UDef-ARP tutorial data can be downloaded from <https://doi.org/10.5281/zenodo.14202000>. Make a folder called Acre. Download the UDef-A data into that folder and extract the files. A subfolder called AcreData will be created where all of your input files will be.

- B. Development of a deforestation risk map starts with the production of forest cover maps for the three dates, T1, T2 and T3. Normally, Verra contracts with a Data Provider to map these based on satellite imagery. Display each of the layers entitled AC_FNF_2010, AC_FNF_2015 and AC_FNF_2020 in whatever GIS you are using. These are the maps for T1, T2 and T3. Note that each indicates the state of the land on the last day of the year indicated (i.e., Dec 31).
- C. Verra also commonly refers to a Forest Cover Benchmark Map (FCBM). Display the layer named AC_FCBM. This was created from the three individual forest cover maps and shows the deforestation that happened in the CAL and CNF periods, which together represent the HRP.

Note that these case study materials vary from typical Verra requirements in two respects. First, Data Providers are normally required to submit a map of Exclusions. Exclusions are areas where deforestation is intentionally ignored. This includes planned deforestation (such as allowed timbering operations) and large-scale infrastructure projects. None have been delineated in these case study data. The second issue is that none of the boundaries of the jurisdiction have been buffered. Normally, Verra requires that data layers extend 10 km into adjacent jurisdictions if the boundary is sub-national. International boundaries are never buffered. In this case study, none of the boundaries are buffered.

EXERCISE 5-2: THE BENCHMARK MODEL: TESTING STAGE / FITTING PHASE

In VT0007, the Benchmark Model is an intentionally simple model that establishes a minimum acceptable level of model skill. It is based on just two layers: 1) a map of distance from the forest edge, and 2) a map of administrative boundaries.

- A. Display the data layers entitled AC_Distance_Forest_Edge_2010, AC_Distance_Forest_Edge_2015 and AC_Distance_Forest_Edge_2020. With the binary forest cover maps we are using, the forest edge is simply the closest non-forest region. Thus these 3 maps were produced using the DISTANCE module and maps of non-forest for each of the 3 years.
- B. Display the map called AC_Municipalities. Municipalities are the administrative regions just below the state level. There are 22 municipalities in Acre.

The Benchmark model uses these two maps to create a set of modeling regions, within which the relative frequency of deforestation is assessed during a historical period (such as the CAL, in this example). First, the map of distance from the forest edge is classified into 30 classes using a geometric progression such that zone 1 is the widest class and is farthest from the forest edge, and zone 30 is the narrowest class right next to the forest edge. The logic here is that the risk of deforestation is higher as you get near to the forest edge. The map of municipalities is then intersected with this 30-class map to produce $30 \times 22 = 660$ modeling regions. These modeling regions are used to assess the relative frequency of deforestation in the historical period. The logic is that, for example, the relative frequency of deforestation in risk zone 30 in one municipality may be quite different from another municipality.

Over time, the modeling regions will change. While the municipalities remain the same, the distance to the forest edge in the CNF will be different. Therefore, using the same logic, a new set of 660 modeling regions will be created for the CNF. The key assumption of the Benchmark Model, is that while the spatial locations of modeling regions may change between the CAL and CNF, the relative frequency of deforestation in the CAL will be the same relative frequency in the CNF in the corresponding (but perhaps spatially different) modeling region. For example, zone 30 in Rio Branco municipality in the CNF may be slightly different spatially than the CAL (because of deforestation in the CAL), but the Benchmark Model assumes that the risk

of deforestation in zone 30 in Rio Branco will remain constant despite the slight change in spatial configuration. It's a simple model that requires minimal data and has proven to produce reasonable results. That said, the ultimate intention is to create an alternative model that is superior to the Benchmark, as we will see.

- C. Open UDef-ARP. On the left side are some large buttons. In the *Testing Stage* section on the top left, click on the *Fitting Phase (CAL)* button. The dialog will change. In the words in the trees above it should say "Testing Stage : Fitting Phase for the Calibration Period (CAL)." Click on the Vulnerability Mapping button. Notice the tabs above the dialog. Keep it as it is, on the *Benchmark Model* tab.
- D. Create a new folder named AcreWorking in your Acre folder. Use the button with three dots on it (the Pick button) to choose your Working Directory (..\Acre\AcreWorking).

The vulnerability map is a 30-class geometric series map of distance zones from the forest edge. It is not a risk map, because it will not be quantified. It only shows relative vulnerability based on this factor alone.

- E. Now fill in the following inputs:

The screenshot shows the UDef-ARP software interface. The title bar reads "UDef-ARP-2_14_test". The main window has a header image of a forest with the text "Testing Stage : Fitting Phase for the Calibration Period (CAL)". On the left, there is a sidebar with four buttons: "1. Vulnerability Mapping", "2. Allocated Risk Mapping", "3. Model Fit Assessment", and "Return to Start Page". The main panel has two tabs: "Benchmark Model" and "Alternative Model". Under the "Benchmark Model" tab, there are several input fields and buttons. The "Working Directory" field contains "D:\Pass\Acre\AcreWorking". The "Distance from forest edge at start of the CAL" field contains "AC_Distance_Forest_Edge_2010.rst". The "Deforestation in the HRP" field contains "AC_Deforestation_HRP_2010_2020.rst". The "Mask of the non-excluded jurisdiction" field contains "AC_Jurisdiction.rst". The "Negligible Risk Threshold (NRT) in the HRP" field is empty. The "Benchmark vulnerability for the CAL" field contains "e.g., Acre_Vulnerability_CAL.tif". There are buttons for "Calculate NRT", "HELP", and "OK".

The vulnerability map is based on distance from the forest edge at the start of the calibration period (CAL), which we looked at earlier. Now display the layer called AC_Jurisdiction. This is a mask image. Areas with 1's are part of the jurisdiction while 0's are not. The dialog specifies a "Mask of the non-excluded jurisdiction, which indicates that if there are any areas to be excluded from consideration, they should also be 0's. In our case, there are no exclusions. Now display the final input which shows all areas that became deforested in the Historical Reference Period (HRP). This will be used to calculate the negligible risk threshold (NRT). The NRT is defined as the distance by which 99.5% of deforestation has occurred over the HRP. Beyond this distance, the vulnerability is effectively zero. Therefore, vulnerability zone 1 starts at the NRT, while zone 30 will be adjacent to the forest edge. Click the CALCULATE button to have it determine the NRT in meters.

1. What is the NRT that it determined for Acre? Write it down for future reference.

Now, for the output file name, specify AC_BM_Vulnerability_CAL.rst. Then click the OK button. (Note: we are using .rst files for this exercise. We could use .tif images/maps and everything would be done the same way. The only difference is that we would probably specify .tif extensions for our output maps, to be consistent.) Since UDef-ARP does have the ability to work with either TIF or TerrSet/Idrisi RST images, you need to specify the extensions for your output files, so it knows which to create.

2. Zoom into parts of the vulnerability image. How would you describe the way the vulnerability zones vary in their width?

- F. Now click on the *Allocated Risk Mapping* button. You will notice that the *Alternative* and *Benchmark Model* tabs disappear. That is because the process for allocating risk is the same for any model, no matter whether it is the Benchmark or an Alternative model. Fill in the input boxes as follows and click OK:

The screenshot shows the UDef-ARP-2_14_test software interface. At the top, there is a header image of a forest with the text "Testing Stage : Fitting Phase for the Calibration Period (CAL)". Below this, there is a sidebar on the left with four buttons: "1. Vulnerability Mapping", "2. Allocated Risk Mapping", "3. Model Fit Assessment", and "Return to Start Page". The main area is titled "Testing Stage : Fitting Phase for the Calibration Period (CAL)" and contains several input fields and buttons. The "Working Directory" is set to "D:\Pass\Acre\AcreWorking". Under "Input Parameters", there are three fields: "Administrative Divisions image" (AC_Municipalities.rst), "Vulnerability image for the CAL (previous step)" (AC_BM_Vulnerability_CAL.rst), and "Deforestation in the CAL" (AC_Deforestation_CAL_2010_2015.rst). Under "Output Parameters", there are three fields: "Modeling Regions in the CAL" (AC_BM_ModReg_CAL.rst), "Relative Frequency Table for the CAL" (AC_BM_ReFreq_CAL.csv), and "Fitted Density in the CAL" (AC_BM_FitDensity_CAL.rst). There are "HELP" and "OK" buttons on the right side.

What this stage will do is to overlay the vulnerability zones you just created with the municipality polygons to create a set of modeling regions. Within each of these regions, the relative frequency of deforestation in the CAL will be calculated and stored in a table (a CSV file). Finally, the fitted density map gives us a spatial representation of our model.

The fitted density map is expressed in hectares (ha) per pixel. The model simply calculates the number of pixels deforested in the CAL within each modeling region. Dividing each sum by the area of its corresponding modeling region (in pixels) yields a relative frequency from 0-1. Multiplying the modeling region relative frequency times the area of pixels ($30\text{m} \times 30\text{m} = 0.09\text{ ha}$) yields the density which is applied to each pixel within its modeling region. In this manner we have gone from what we know (only

the exact pixels which deforested in the CAL) to a general mapping of relative frequency for all pixels in Acre. It is a generalization in a similar sense as a regression line is a generalization of a scatter of points in an X/Y graph.

3. If you sum all of the fitted density pixels in Acre, the total will equal the area of the features in another map in your Acre data set. What map is that?

- G. Now that we have fitted a model to the deforestation in the CAL, how good a model is it? Click on the *Model Fit Assessment* button, then fill out the dialog with the following input and output names. Note that the Output Plot Filename can be a PNG file or a JPG file. Now click OK to create the assessment.

To assess performance, UDef-ARP sets up a coarse grid with cells equal to the median size of REDD projects in the jurisdiction (or 100,000 ha if this is unknown). In Acre, a search of Verra's database (registry.verra.org) indicated a median size of approximately 42,000 hectares. For each grid cell that falls completely within the jurisdiction boundary, it then calculates the sum of fitted densities (which equals the fitted area of deforestation) as well as the corresponding actual area of deforestation. Using the actual deforestation as the X variable and the fitted area as the Y variable, an X/Y plot (also known as a *Calibration Plot* or *Actual-vs-Predicted Chart*¹) illustrates the quality of the fit.

In the performance chart, each dot represents one of the evaluation grid cells. If the model were a perfect fit, the dots would all line up along the 1:1 line (the dashed red line). Clearly, they don't. The green

¹ Some sources use X to indicate the predicted value and Y the actual value. Here we maintain consistence with the observed values being designated as X.

line is the OLS (Ordinary Least Squares) best fit, while the yellow line is an alternative regression fit (Theil-Sen) that is more resistant to outliers. We will concentrate on the OLS fit. In a performance chart such as this, the slope of the OLS line indicates the accuracy of the model, where a slope of 1 (45 degrees) is perfect accuracy². The scatter of dots from the OLS line indicates the precision of the model, and is summarized by the R^2 . If the dots all lined up with the OLS line, the precision would be perfect. Note that it is possible to have a model with high accuracy and low precision and vice versa.

In the case of Acre during the CAL, there is a substantial downward bias in the OLS fit relative to the 1:1 line. There are several reasons for this bias, including the following:

- Errors in the forest cover and derived deforestation maps are a substantial source of negative (downward) bias in the slope of the regression. Linear regression assumes that the independent variable (X) is without error. However, with map error, $X_{\text{measured}} = X_{\text{true}} + \text{noise}$. The slope of the regression is calculated as the ratio of the covariance between X and Y to the variance in X. The covariance contains this noise in the contribution from X, but the variance contains this noise effect squared, which increases the denominator and thus the slope becomes smaller.
- Model error is another substantial source. A model is comprised of both data and a method to establish a structural relationship between the data and deforestation. The Benchmark model is intentionally simple. It is based on only two variables (distance from the forest edge and administrative divisions), and sets a minimum standard of model skill. Risk map providers are required to develop alternative models that meet or exceed this standard. Note that the primary source of the bias here is the fact that only two covariates are used.
- Non-stationarity (inconsistency in the nature of deforestation) over space makes it difficult for the empirical methodology used to establish a consistent relationship between deforestation and the covariates.
- Regression also assumes that the errors are distributed normally and identically across the range of values tested (a property known as homoscedasticity). However, this cannot be true here. Since probabilities cannot be negative, errors in Y near 0 are skewed in the positive direction. This promotes a negative slope and a positive intercept at $X=0$.

In addition to a negative slope indicating imperfect accuracy, the scatter of dots evident in the Acre Benchmark fit reflects imprecision. The sources of this are many, including imprecision in the explanatory covariates and map error. **Ultimately, the goal is to develop a model that yields an OLS fit with minimal scatter (i.e., a high R^2) and a slope near 1.0.**

Since the 1:1 line indicates the position of a perfect fit, the error associated with any of the evaluation grid cells can be measured as the absolute vertical distance between its dot on the graph and the 1:1 line. The Median Absolute Error (MedAE) between the fitted densities and the actual densities among the 328 grid cells is 134.47 ha, which represents 0.30% of the 42,000 ha area of these cells. This is the Benchmark metric for the fit in the CAL. **Any alternative model that is used must demonstrate a fit with a lower (better) MedAE than this.**

² This is not true of all regression models. However, it is true here because the total predicted quantity over the whole map is forced to equal the actual quantity of deforestation (which it knows from the deforestation map for the CAL).

Along with the performance chart, UDef-ARP also produces a residual plot. To understand better how this was made, in a GIS (e.g. QGIS), we will try to visualize the residuals better. Display the residuals image and add a raster layer named AC_Reverse_Mask on top of the residuals image. The reverse mask has a value of 0 (background) in the area of the Acre jurisdiction.

If using QGIS, in the *Layers Panel*, right-click on AC_Reverse_Mask, then choose *Properties - Transparency*. Under *Additional no data value* or *Transparent pixel list*, Add value 0, and click Apply. This will make the pixels with value 0 become transparent.

Then apply something similar to a symmetrical stretch to make it easier to appreciate the nature of the residuals. In QGIS' *Layers panel*, right-click on AC_BM_Fit_CAL_Residuals, then choose *Properties - Symbology*. Choose *Render type - Singleband pseudocolor*. Use a diverging *Color ramp* such as RdBu (under All Color Ramps), Spectral, or Blue-White-Red, and also change the *Max* value on that same *Band Rendering* section to 1822.38, to match the *Min*, which will set the range of values symmetrically around zero. This is essentially what a symmetrical stretch would do in TerrSet's display.

To understand where the residuals are happening, add a third layer to your composition. Add the AC_FCBM layer. You can now compare the two maps by clicking the visibility checkbox for AC_FCBM in the Layers Panel on and off. This lets you determine where high residuals occur, whether they correspond to forest/non-forest boundaries, and whether errors cluster in particular land-cover classes.

4. Where do you see that the Benchmark tends to fit values of deforestation that are too low? And where does it tend to predict too high?

EXERCISE 5-3: THE BENCHMARK MODEL: TESTING STAGE / PREDICTION PHASE

The fit indicates how well a model fits (explains) the data it was trained on (the deforestation in the CAL relative to the covariates used). However, it does not say how well it will perform in a future prediction. This is where the data for the CNF comes in.

- A. In UDef-ARP, click the *Return to Start Page* button, then within the *Testing Stage*, click the *Prediction Phase (CNF)* button. Then click the *Vulnerability Mapping* button, and verify that the active tab is the *Benchmark Model*. Then fill out the dialog as follows and click the OK button.



Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Benchmark Model		Alternative Model	
Testing Stage : Prediction Phase for the Confirmation Period (CNF)			
Working Directory	D:\Pass\Acre\AcreWorking		...
Input Parameters			
Distance from forest edge at start of the CNF	AC_Distance_Forest_Edge_2015.rst		...
Mask of the non-excluded jurisdiction	AC_Jurisdiction.rst		...
Negligible Risk Threshold (NRT) in the HRP	3405		
Output Parameters			
Benchmark vulnerability for the CNF	AC_BM_Vulnerability_CNF.rst		
			HELP OK

1. Vulnerability Mapping

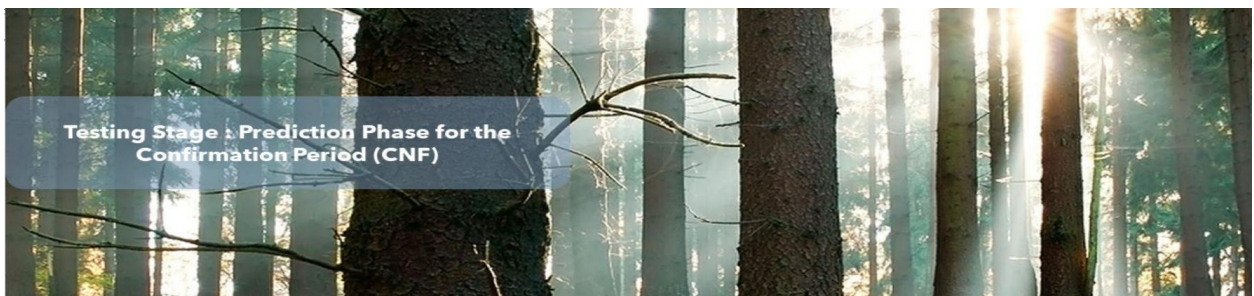
2. Allocated Risk Mapping

3. Model Prediction Assessment

Return to Start Page

This will create a new 30-class vulnerability map for the CNF. Note that the Negligible Risk Threshold (NRT) is set to be the same as that used before (3405 m). This is because it was calculated for the whole HRP which includes both the CAL and the CNF.

- B. Now that the vulnerability map has been created for the CNF, the next step is to allocate the risk. Click on the *Allocated Risk Mapping* button. Then fill out the dialog as follows and click the OK button.



Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Benchmark Model		Alternative Model	
Testing Stage : Prediction Phase for the Confirmation Period (CNF)			
Working Directory	D:\Pass\Acre\AcreWorking		...
Input Parameters			
Administrative Divisions Image	AC_Municipalities.rst		...
CAL Relative Frequency Table	AC_BM_RelFreq_CAL.csv		...
Vulnerability for the CNF (previous step)	AC_BM_Vulnerability_CNF.rst		...
Deforestation in the CNF	AC_Deforestation_CNF_2015_2020.rst		...
Maximum Iterations for Solution Convergence	3		
Output Parameters			
Modeling Region in the CNF	AC_BM_ModReg_CNF.rst		
Predicted Density in CNF	AC_BM_Predicted_Density_CNF.rst		
			HELP OK

1. Vulnerability Mapping

2. Allocated Risk Mapping

3. Model Prediction Assessment

Return to Start Page

With this procedure, a new set of modeling regions will be created by intersecting the 30 vulnerability zones with the municipality polygons. When the model was fit, these modeling regions were used to calculate the relative frequency of deforestation within each. In this phase, however, each modeling

region is assigned the relative frequencies calculated in the CAL. Each modeling region has an ID associated with its definition. For example, modeling region 30142 represents vulnerability zone 30 (right next to non-forest) in municipality 142. In the CNF, the same modeling region codes are used even though their positions may have changed. For example, municipality 142 will have experienced new deforestation in the CAL, so vulnerability zone 30 in municipality 142 will be in a slightly different position.

1. In meteorology, a persistence forecast says that tomorrow's temperature will be exactly like today. In what way is the forecasting logic of UDef-ARP similar? In what way is it different?

You may have noticed that there is a parameter for the maximum number of iterations for solution convergence. By design, the prediction is forced to have the same quantity that is known to have deforested in the CNF. This is so that the evaluation of prediction skill focuses only on the ability of the model to predict where deforestation will happen. A similar thing will happen when the model is ultimately applied, as will be explained later. In matching the correct quantity, it may happen that individual pixels may be allocated more than 0.09 ha (the area of individual pixels). In this case, the software sets those pixels to be 0.09 and increases the remaining pixels to compensate. The algorithm used converges very quickly, which is why 3 iterations are usually sufficient.

2. If you sum all of the predicted density pixels in Acre, what should it total?

- C. Click on the *Model Prediction Assessment* button. Then fill out the dialog as follows and click the OK button.

Testing Stage - Prediction Phase for the Confirmation Period (CNF)

Working Directory: D:\Pass\Acre\AcreWorking

Input Parameters

Adjusted Predicted Density in the CNF	AC_BM_Predicted_Density_CNf.rst
Mask of the Jurisdiction	AC_Jurisdiction.rst
Deforestation in the CNF	AC_Deforestation_CNf_2015_2020.rst
Deforestation in the CAL	AC_Deforestation_CAL_2010_2015.rst
Forest Areas at start of CAL	AC_forest_2010.rst
Assessment Grid Cell Area or 100,000 (ha)	42000

Output Parameters

Title of Plot	Acre Benchmark Prediction CNF
Output plot filename	AC_BM_Prediction_CNf.png
Output residual image	AC_BM_Prediction_CNf_Residuals.rst
Output combined deforestation review image	AC_BM_Prediction_CNf_Review.rst
Set X-Axis Max	Default
Set Y-Axis Max	Default

Buttons: 1. Vulnerability Mapping, 2. Allocated Risk Mapping, 3. Model Prediction Assessment, Return to Start Page, HELP, OK

3. Notice that while the prediction model skill is quite similar to the fit, the slope is slightly lower, the R^2 is slightly worse and the MedAE is substantially higher (worse). What factor(s) do you think might have caused this?

EXERCISE 5-4: ALTERNATIVE MODELS: TESTING STAGE / FITTING PHASE

The Benchmark model is intentionally simple. It sets a minimum standard of skill. The goal of UDef-ARP is actually to create an alternative model that exceeds this quality.

An empirical alternative model is a deforestation risk model developed independently of the benchmark model using historical observations of deforestation and one or more spatial predictor variables that are believed to influence future forest loss. These models are typically created by analyzing the relationship between past deforestation and factors such as accessibility, proximity to roads or settlements, topography, land tenure, or other biophysical and socioeconomic drivers. The result is a spatially explicit risk surface that estimates the relative likelihood of future deforestation across the jurisdiction. Within the UDef-ARP framework, empirical alternative models provide an independent representation of deforestation risk that can be compared against the benchmark model. By evaluating multiple plausible risk models, users can identify the model that most accurately predicts observed deforestation patterns and therefore provides the most appropriate basis for spatial allocation.

Alternative models may be developed using a variety of software platforms and modeling approaches, provided that they produce a continuous, spatially explicit surface representing relative deforestation risk. Common options include:

- TerrSet Land Change Modeler (LCM), including Multi-Layer Perceptron (MLP) and logistic regression approaches.
- ArcGIS Pro or ArcMap using spatial statistics, regression, machine learning, or suitability modeling workflows.
- QGIS using native processing tools and external plugins for statistical or machine learning analysis.
- Open-source programming environments such as Python (e.g., scikit-learn, XGBoost, TensorFlow, PyTorch) or R (e.g., randomForest, ranger, caret, terra).
- Other custom modeling frameworks that incorporate historical deforestation observations and spatial predictor variables to generate a relative deforestation risk surface.

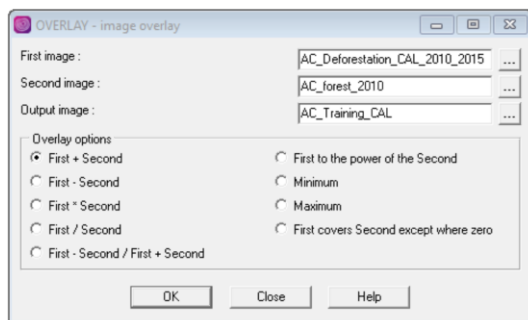
In this exercise, we will use TerrSet to create a MultiLayer Perceptron (MLP) alternative model and compare it to the Benchmark. To do so, we will develop a MultiLayer Perceptron (MLP) model as an alternative to Distance from the Forest Edge as an expression of vulnerability to deforestation. The MLP model will in turn be based on discovering the relationship between deforestation and a set of covariates – variables that are believed to spatially and temporally covary with deforestation.

- A. The covariates for this set of data are distance from pasture, urban, rivers and roads for each time period; elevation; protected_areas_relF and municipalities_relF. Display the various covariate layers.

Most of the covariates are distance maps. Remember that each of the dates indicate the last day of the year. Thus, a distance map labeled as 2010, for example, represents the distance as of Dec 31, 2010. It could equally be thought of as Jan 1, 2011, the start of the CAL. Most of these layers were developed from land cover data provided by MapBiomias Brazil Collection 9 (brasil.mapbiomas.org). The exceptions were the rivers and roads data from OpenStreetMap, downloaded from Geofabrik.de, Elevation data from NASADEM (<https://www.earthdata.nasa.gov/data/catalog/lpcloud-nasadem-hgt-001>), municipalities downloaded from GADM.org and WDPA protected areas downloaded from www.protectedplanet.net/en/thematic-areas/wdpa?tab=WDPA.

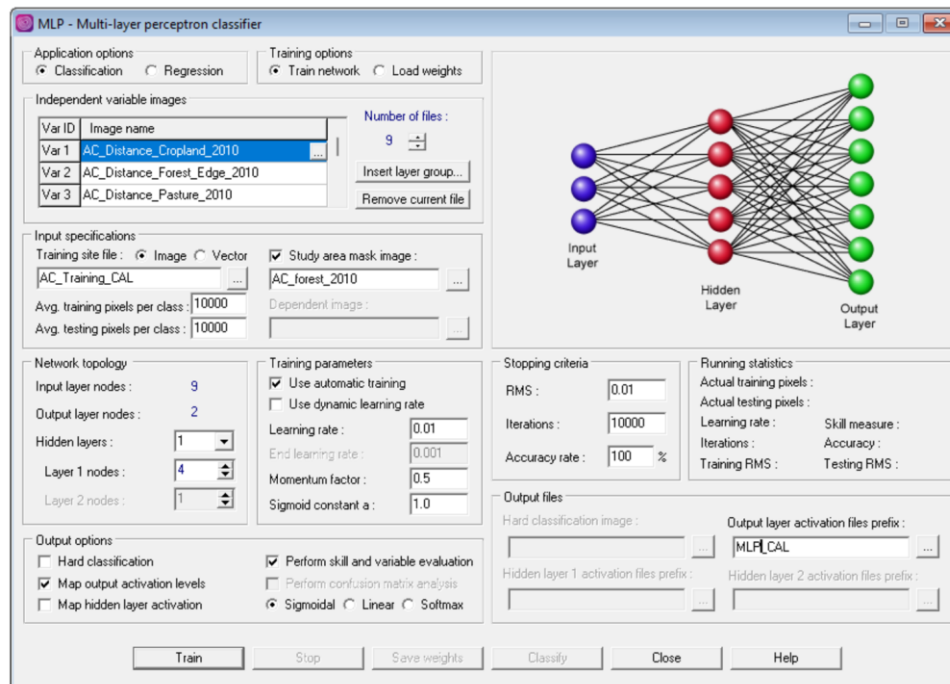
The two images with RelF in the name contain information about the relative frequency of deforestation for the date range indicated (again, referencing Dec 31). For the CAL, since this is for model development, MLP will have information on the actual location of deforestation as well as the relative frequency of deforestation within municipalities and protected areas (and thus also, unprotected areas).

- B. To run MLP, you are going to need a training image that indicates areas that deforested in the CAL and areas that persisted as forest. To do this, run OVERLAY as follows:

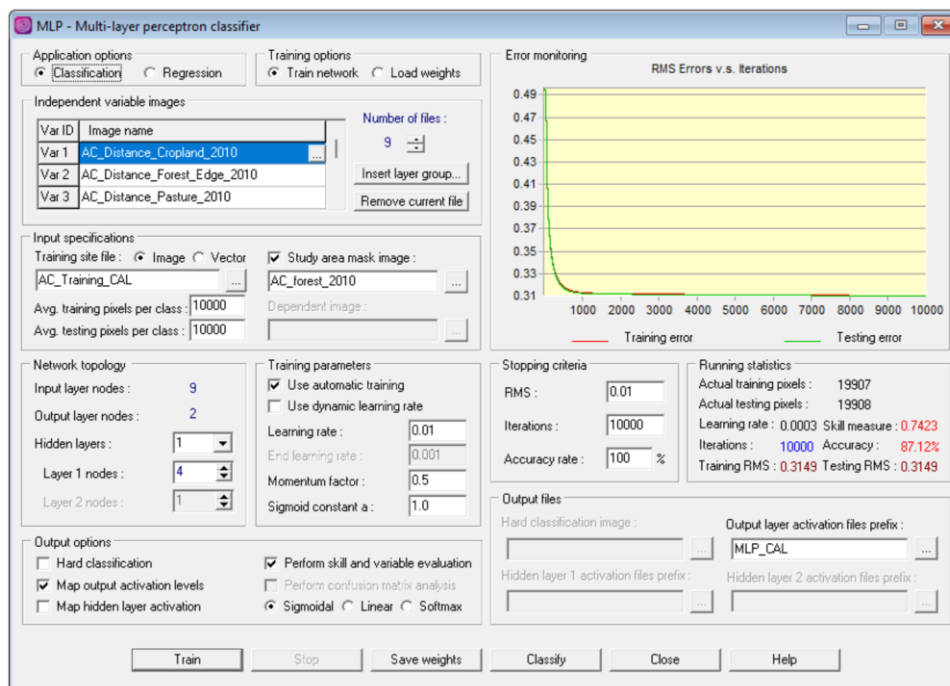


This will create an image where 0's are non-forest, 1's are forest persistence and 2's are deforestation in the CAL.

- C. Run MLP and click the INSERT LAYER GROUP button and specify the AC_Covariates_CAL group file. This will specify 9 covariates. Then fill out the rest of the dialog to look as follows:



Be careful to make sure that every part of the dialog is correctly specified. Then click TRAIN. When it finishes, your dialog should look similar (but probably not identical) to this:



Then click the SAVE WEIGHTS button and specify AC_CAL as the name for the weights file. Now click the CLASSIFY button. When it has finished it will display an image named MLP_CAL_1 along with a report (don't close the report until indicated to do so below). By default, MLP always displays the output

associated with the first class. This is the transition potential for forest persistence, which is not of interest. Display its companion image, MLP_CAL_2. This is the transition potential for deforestation.

1. Looking at the MLP report, what was the overall skill? What was the skill for predicting deforestation (class 2)?
2. Skill is a measure from -1 to +1. A value of 0 means skill no better than random. Generally, a skill measure of 0.5 or greater is good. Skill is a better measure than accuracy because accuracy includes two components – change accuracy and proficient accuracy. What do you think the chance accuracy is in this case? Why?
3. Look at the last section of the report (Backward Stepwise Constant Forcing). This is one of the most useful components of the report. The importance of covariates to the model can be determined from the bottom up. What are the three most important covariates to the model (look at Step 6)?

- D. Now return to UDef-ARP's main page. Within the *Testing Stage*, click the *Fitting Phase* button. Select *Vulnerability Mapping* button and choose the tab for *Alternative Model*. Then fill out the dialog as follows and then click OK.

Testing Stage : Fitting Phase for the Calibration Period (CAL)

Benchmark Model **Alternative Model**

Working Directory: D:\Pass\Acre\AcreWorking

Empirical Transition Potential in the CAL: MLP_CAL_2.rst

Mask of the non-excluded jurisdiction: AC_Jurisdiction.rst

Forest Areas at start of CAL: AC_forest_2010.rst

Alternative vulnerability for the CAL: AC_MLP_Vulnerability_CAL.rst

HELP OK

4. How similar is the vulnerability map based on the MLP transition potential (AC_MLP_Vulnerability_CAL) to that produced by the Benchmark (AC_BM_Vulnerability_CAL)? List at least two differences you see.
- E. The remaining steps for the alternative risk mapping in the CAL are the same as we did for the Benchmark. Click the *Allocated Risk Mapping* button (within the Testing Stage, Fitting Phase) and fill out the dialog as follows:



1. Vulnerability Mapping
2. Allocated Risk Mapping
3. Model Fit Assessment
Return to Start Page

Testing Stage : Fitting Phase for the Calibration Period (CAL)

Working Directory
D:\Pass\Acre\AcreWorking

Input Parameters
Administrative Divisions image
AC_Municipalities.rst
Vulnerability image for the CAL (previous step)
AC_MLP_Vulnerability_CAL.rst
Deforestation in the CAL
AC_Deforestation_CAL_2010_2015.rst

Output Parameters
Modeling Regions in the CAL
AC_MLP_ModReg_CAL.rst
Relative Frequency Table for the CAL
AC_MLP_RelFreq_CAL.csv
Fitted Density in the CAL
AC_MLP_FL_Density_CAL.rst

HELP
OK

- F. Now that the MLP-based density map for the CAL is complete, choose *Model Fit Assessment* to see how it did.



1. Vulnerability Mapping
2. Allocated Risk Mapping
3. Model Fit Assessment
Return to Start Page

Testing Stage : Fitting Phase for the Calibration Period (CAL)

Working Directory
D:\Pass\Acre\AcreWorking

Input Parameters
Fitted Density for the CAL image (ha/pixel)
AC_MLP_FL_Density_CAL.rst
Mask of the Jurisdiction
AC_Jurisdiction.rst
Deforestation in the CAL image
AC_Deforestation_CAL_2010_2015.rst
Assessment Grid Cell Area or 100,000 (ha)
42000

Output Parameters
Title of Plot
Acre MLP Fit CAL
Output plot filename
AC_MLP_FL_CAL.png
Output residual image
AC_MLP_FL_CAL_Residuals.rst
Set X-Axis Max manually
Default
Set Y-Axis Max manually
Default

HELP
OK

5. What were the R^2 , OLS slope and MedAE values for the MLP fit? How does this compare with the Benchmark?

EXERCISE 5-5: ALTERNATIVE MODELS: TESTING STAGE / PREDICTION PHASE

Looking at the goodness of fit of a model is an important metric, but it doesn't ensure that the model will perform well when it really has to make a true prediction into the future. It sometimes happens that a model will fit the data extremely well, but then not so do well with a future prediction. The reasons are primarily two-fold:

- First, it's possible that the model overfits the data. When the data contains errors, some models are actually pretty good at fitting the errors as well as the good data. Then when the model is tested, it turns out that it performs poorly.
- Second, it could relate to temporal non-stationarity. The future doesn't always behave like the past. This is very common. Map error is also present, but in this context, would appear as non-stationarity (since we don't know if it's error or not).

To assess the predictive skill of our MLP model, we will use it to predict the deforestation in the CNF and then assess how well it did.

- A. Open the MLP module in TerrSet. At the top of the dialog, change the training option to be *Load Weights*. Just below that, click the Insert Layer Group button and select the AC_COVARIATES_CNF group file³. This group file is the same as for the CAL, except that any dynamic variables (variables which change over time) have been updated to reflect the beginning of the CNF. This includes all of the distance covariates which are updated to 2015, and the two relative frequency covariates (municipalities and protected areas).

Now ensure that the rest of the dialog is filled out as follows, and click the CLASSIFY button:

³ Note: if the MLP dialog was still open from the previous run and had the covariates for the CAL already entered into the grid, clear them from the dialog by clicking on "Remove current file" for each of them. Then add the group file for the dynamic variables for the CNF.

MLP - Multi-layer perceptron classifier

Application options: ☒ Classification ☐ Regression

Training options: ☐ Train network ☒ Load weights

Independent variable images

Var ID	Image name
Var 1	AC_Distance_Cropland_2010
Var 2	AC_Distance_Forest_Edge_2010
Var 3	AC_Distance_Pasture_2010

Number of files : 9

Insert layer group...

Remove current file

☒ Study area mask image : AC_forest_2015

Weight file name : AC_CAL

Output options

☐ Hard classification ☒ Perform skill and variable evaluation

☒ Map output activation levels ☐ Perform confusion matrix analysis

☐ Map hidden layer activation ☒ Sigmoidal ☐ Linear ☐ Softmax

Output files

Hard classification image : ...

Output layer activation files prefix : MLP_CNF

Hidden layer 1 activation files prefix : ...

Hidden layer 2 activation files prefix : ...

Classify Close Help

There is no training stage here. The model is exactly the same as for that CAL – you entered the weight file previously created. The only thing that's changes is the updating of the covariates. As before, the result of interest is the MLP_CNF_2 image.

- B. Now create the MLP-based vulnerability map for the CNF. Return to UDef-ARP's start page and choose *Prediction Phase* from the *Testing Stage*. Choose the *Alternative Model* tab, then input the following:



Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Benchmark Model
Alternative Model

1. Vulnerability Mapping

2. Allocated Risk Mapping

3. Model Prediction Assessment

Return to Start Page

Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Working Directory

...

Input Parameters

Empirical Transition Potential in the CNF

...

Mask of the non-excluded jurisdiction

...

Forest Areas at start of CNF

...

Output Parameters

Alternative vulnerability for the CNF

HELP
OK

C. Then allocate the MLP-based risk for the CNF. Here are the inputs:



Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Benchmark Model
Alternative Model

1. Vulnerability Mapping

2. Allocated Risk Mapping

3. Model Prediction Assessment

Return to Start Page

Testing Stage : Prediction Phase for the Confirmation Period (CNF)

Working Directory

...

Input Parameters

Administrative Divisions Image

...

CAL Relative Frequency Table

...

Vulnerability for the CNF (previous step)

...

Deforestation in the CNF

...

Maximum Iteration for Solution Convergence

Output Parameters

Modeling Region in the CNF

Predicted Density in CNF

HELP
OK

D. Finally, create an assessment of the prediction for the CNF. Note that this dialog asks for inputs for both the CNF and the CAL. Also note the new output – the review map. This is not much of interest because it will be identical to the FCBM image we already have. It's provided simply for those who do not have such an image to start.

1. What were the R^2 , OLS slope and MedAE values for the MLP prediction? How does this compare with the Benchmark?
2. Based on this information, which is the better model? The Benchmark or MLP?

You will have noticed that both models performed worse in the CNF than the CAL. This is normal. Neither the Benchmark nor MLP has a tendency to overfit, so the difference is probably mostly the result of non-stationarity.

EXERCISE 6-5: CHALLENGE 1: APPLICATION STAGE / FITTING PHASE

The Application Stage proceeds exactly like the Testing Stage. Therefore, your challenge is to create a new model for the entire HRP using MLP as an alternative model (since you figured out that this was best). Note that for training your MLP model you should use the AC_Covariates_HRP group file. Remember that the beginning of the HRP is the same date as the beginning of the CAL.

1. How did your MLP model for the HRP compare to that in the CAL. These were both fitting phases – only the length of time differed.

Note: if you get a fitted density image that is all zeroes, then your input images didn't make sense (they may have been based off the wrong time period).

EXERCISE 6-5: CHALLENGE 2: APPLICATION STAGE / PREDICTION PHASE

Now that you have a fitted MLP model for the HRP, create a prediction for the BVP (VP). For this you will be asked for the expected annual deforestation (in hectares per year) for the BVP. This is because Verra requires that an estimate of expected deforestation activity into the future be based on high-resolution imagery sampling. For the BVP in this tutorial, you can use the figure 46,304 ha/year.